

IP возможные

172.16.0.0/10.0.0.0/192.168.0.0/100.64.0.0/10.64.0.0

Cat /etc/passwd

ip -c -br a

SHOW INTERFACE BRIEF

1. Таблица IP-адресов и интерфейсов

Устройство	Интерфейс	IP-адрес	Назначение
ISP	ens3	DHCP/WAN	Интернет
ISP	ens4	172.16.1.1/28	Сеть HQ
ISP	ens5	172.16.2.1/28	Сеть BR
HQ-RTR	isp	172.16.1.2/28	Подключение к ISP
HQ-RTR	vl100	10.10.100.1/27	HQ-SRV VLAN100
HQ-RTR	vl200	10.10.200.1/27	HQ-CLI VLAN200
HQ-RTR	vl999	10.10.30.1/29	Management VLAN999
HQ-RTR	tunnel.0	10.10.10.1/30	GRE Tunnel
HQ-SRV	ens3	10.10.100.2/27	DNS/BIND9
HQ-CLI	ens3	DHCP	Клиент VLAN200
BR-RTR	isp	172.16.2.2/28	Подключение к ISP
BR-RTR	br	10.20.20.1/28	LAN BR
BR-RTR	tunnel.0	10.10.10.2/30	GRE Tunnel
BR-SRV	ens3	10.20.20.2/28	Сервер BR

2. ISP

```
hostnamectl set-hostname isp
exec Bash

# WAN (магистраль)
mcedit /etc/net/ifaces/ens3/options 192.168.31.141/24
TYPE=eth
BOOTPROTO=dhcp

# HQ
mkdir -p /etc/net/ifaces/ens4
mcedit /etc/net/ifaces/ens4/ipv4address "172.16.1.1/28"
mcedit /etc/net/ifaces/ens4/options
TYPE=eth
BOOTPROTO=static

# BR
mkdir -p /etc/net/ifaces/ens5
mcedit /etc/net/ifaces/ens5/ipv4address
mcedit /etc/net/ifaces/ens5/options
TYPE=eth
BOOTPROTO=static

# IP Forward
mcedit /etc/net/sysctl.conf
net.ipv4.ip_forward = 1

systemctl restart network

iptables -t nat -A POSTROUTING -o ens3 -j MASQUERADE

iptables-save > /etc/sysconfig/iptables
systemctl enable --now iptables

timedatectl set-timezone Europe/Moscow
```

3. HQ-RTR

```
enable
configure terminal
Ip route 0.0.0.0/0 172.16.1.1
hostname HQ-RTR
ip domain-name au-team.irpo

username net_admin
password P@ssw0rd
role admin

interface isp
ip address 172.16.1.2/28
ip nat outside
exit

interface vl100
ip address 10.10.100.1/27
ip nat inside
exit

interface vl200
ip address 10.10.200.1/27
ip nat inside
exit

interface vl999
ip address 10.10.30.1/29
ip nat inside
exit

port gel
service-instance gel/vl100
encapsulation dot1q 100 exact
rewrite pop 1
connect ip interface vl100
exit
service-instance gel/vl200
encapsulation dot1q 200 exact
rewrite pop 1
connect ip interface vl200
exit
service-instance gel/vl999
encapsulation dot1q 999 exact
rewrite pop 1
connect ip interface vl999
exit

interface tunnel.0
description GRE-to-br-rtr
```

```

ip address 10.10.10.1/30

ip tunnel 172.16.1.2 172.16.2.2
mode gre

ip ospf authentication message-
digest

ip ospf message-digest-key 1 md5
P@ssw0rd exit

router ospf 1

ospf router-id 10.10.10.1
passive-interface default

no passive-interface tunnel.0
network 10.10.100.0/27 area 0
network 10.10.200.0/27 area 0
network 10.10.30.0/29 area 0
network 10.10.10.0/30 area 0
exit

ip nat pool HQ 10.10.30.0-
10.10.200.255

ip nat source dynamic inside-
to-outside pool HQ overload
interface ISP

no dhcp-server 1

ip pool VLAN200 10.10.200.2-10.10.200.30

dhcp-server 1

pool VLAN200 1

mask 255.255.255.224
gateway 10.10.200.1

dns 10.10.100.2

domain-name au-
team.irpo

interface vl200

dhcp-server 1

exit

ntp timezone utc+3
write memory

```

4. HQ-SRV (DNS/BIND9)

```
hostnamectl set-hostname hq-srv.au-team.irpo
```

```
mkdir -p /etc/net/ifaces/ens3
```

```
mcedit /etc/net/ifaces/ens3/ipv4address 10.10.100.2/27
```

```
mcedit /etc/net/ifaces/ens3/ipv4route default via 10.10.100.1
```

```
mcedit /etc/net/ifaces/ens3/options
```

```
TYPE=eth
```

```
BOOTPROTO=static
```

```
systemctl restart network
```

```
mcedit /etc/net/ifaces/ens3/resolv.conf search
```

```
au-team.irpo
```

```
mcedit /etc/net/ifaces/ens3/resolv.conf
```

```
nameserver 127.0.0.1
```

```
useradd sshuser -u 2026
```

```
passwd sshuser
```

```
P@ssw0rd вписать
```

```
gpasswd -a sshuser wheel
```

```
mcedit /etc/sudoers
```

```
%wheel ALL=(ALL:ALL) NOPASSWD: ALL
```

```
mcedit /etc/openssh/sshd_config
```

```
Port 2026
```

```
MaxAuthTries 2
```

```
Banner /etc/openssh/banner
```

```
AllowUsers sshuser
```

```
mcedit /etc/openssh/banner Authorized access only
```

```
systemctl restart sshd
```

apt-get install bind bind-utils -y пакете думаю не

надо, установлено и так все

```
// -коммент (удалить)
```

```

mcedit /var/lib/bind/etc/options.conf
listen-on port 53 { any; }; forwarders {
77.88.8.8; }; allow-query { any; };

mcedit /var/lib/bind/etc/rfc1912.conf
zone "au-team.irpo" { type master;
file "au-team.irpo";

};

zone "100.10.10.in-addr.arpa" { type
master;
file "10.10.100.in-addr.arpa";
};

zone "200.10.10.in-addr.arpa" { type
master;
file "10.10.200.in-addr.arpa";
};

cd /var/lib/bind/etc/zone

cp empty au-team.irpo

cp empty 10.10.100.in-addr.arpa
cp empty 10.10.200.in-addr.arpa

# ПРЯМАЯ ЗОНА
mcedit au-team.irpo

$TTL 1D

@ IN SOA hq-srv.au-team.irpo. root.au-team.irpo. (

2026051001 12H 1H 1W 1H )

@          IN NS          hq-srv.au-team.irpo.

hq-rtr IN A10.10.100.1

br-rtr IN A10.20.20.1

hq-srv IN A10.10.100.2

hq-cli IN A10.10.200.2

br-srv IN A10.20.20.2

docker IN A172.16.1.1

web  IN A   172.16.2.1

# ОБРАТНАЯ VLAN100

```

```
mcedit 10.10.100.in-addr.arpa
1 IN PTR hq-rtr.au-team.irpo.

2 IN PTR hq-srv.au-team.irpo.

# ОБПATHАЯ VLAN200

mcedit 10.10.200.in-addr.arpa
1 IN PTR hq-rtr.au-team.irpo.
2 IN PTR hq-cli.au-team.irpo.

rndc-confgen > /etc/bind/rndc.key

sed -i '6,$d' /etc/bind/rndc.key

chown -R named:named
/var/lib/bind/etc/zone/*

named-checkconf -z

systemctl enable --now
bind

mcedit /etc/net/ifaces/ens3/resolv.conf search
au-team.irpo

mcedit /etc/net/ifaces/ens3/resolv.conf
nameserver 127.0.0.1

timedatectl set-timezone Europe/Moscow
```

5. HQ-CLI

```
hostnamectl set-hostname hq-cli.au-team.irpo

mkdir -p /etc/net/ifaces/ens3

mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=dhcp
```

```
systemctl restart network
timedatectl set-timezone Europe/Moscow
```

6. BR-RTR

```
enable
configure terminal

hostname BR-RTR
ip domain-name au-team.irpo

username net_admin
password P@ssw0rd
role admin

interface isp
 ip address 172.16.2.2/28
 ip nat outside
 exit

interface br
 ip address 10.20.20.1/28
 ip nat inside
 exit

port tel
 service-instance tel/br
 encapsulation untagged
 connect ip interface br
 exit

interface tunnel.0
 description GRE-to-HQ-RTR
 ip address 10.10.10.2/30
 ip tunnel 172.16.2.2 172.16.1.2 mode gre
 ip ospf authentication message-digest
 ip ospf message-digest-key 1 md5 P@ssw0rd
 exit

router ospf 1
 ospf router-id 10.10.10.2
 passive-interface default
 no passive-interface tunnel.0
 network 10.20.20.0/28 area 0
 network 10.10.10.0/30 area 0
 exit
Ip nat pool br 10.20.20.1-10.20.20.64
ip nat source dynamic inside-to-outside pool BR-POOL overload interface isp
```

```
ip route 0.0.0.0/0 172.16.2.1

ntp timezone utc+3
write memory
```

7. BR-SRV

```
hostnamectl set-hostname br-srv.au-team.irpo

mkdir -p /etc/net/ifaces/ens3
        mcedit /etc/net/ifaces/ens3/ipv4address 10.20.20.2/28
        mcedit /etc/net/ifaces/ens3/ipv4route default via 10.20.20.1

mcedit /etc/net/ifaces/ens3/options
TYPE=eth
BOOTPROTO=Static (Преднастрой)

useradd sshuser -u 2026
        | passwd sshuser, и пиши пароль сука P@ssw0rd
gpasswd -a sshuser wheel

mcedit /etc/net/ifaces/ens3/resolv.conf search au-team.irpo
mcedit /etc/net/ifaces/ens3/resolv.conf nameserver 10.10.100.2

systemctl restart network
timedatectl set-timezone Europe/Moscow SSH АДМИН???
```